All Five DNA Letters Found in an Asteroid

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Scientists have found all five "letters" of DNA in a very old **asteroid**. The letters are tiny **molecules** called **nucleobases**. They spell out the **instructions** for life in DNA and RNA. The five are adenine, guanine, cytosine, thymine and **uracil**. A team of Japanese scientists found them in rocks from the **asteroid** Ryugu. Japan's Hayabusa2 **spacecraft** brought the rocks back to Earth in 2020. Scientists had found one of the five letters, **uracil**, in the same rocks in 2023. Now they have found all five. Rocks that fall to Earth can pick up dirt from our planet. But the Ryugu rocks are clean. Hayabusa2 brought back 5.4 grams of rock that was kept safe from Earth's dirt. The scientists studied the rocks in super-clean labs. The find suggests that the building blocks of life may have formed in space long ago. **Asteroids** like Ryugu are left over from the early days of our Solar System, about 4.5 billion years ago. They may have carried these building blocks to the young Earth.

Word Treasure

nucleobases -- Nucleobases are the tiny molecules that spell out genetic instructions.
asteroid -- An asteroid is a rock that flies around the Sun.
spacecraft -- A spacecraft is a vehicle that travels in space.
molecules -- Molecules are tiny bits of matter that make up everything.
uracil -- Uracil is one of the five nucleobases, the letters of DNA and RNA.
instructions -- Instructions tell you how to do something.

Background you need

DNA and RNA hold the instructions that tell living things how to grow.

Asteroids are old space rocks left over from the early Solar System.

Break it down

WHO: Japanese scientists and Japan's Hayabusa2 spacecraft
WHAT: All five DNA letters found in an asteroid
WHERE: Samples from the asteroid Ryugu
WHY: To learn if life's building blocks came from space

Why it matters

Learning where life's building blocks came from helps us understand where we came from, too.

Test what you remember

1. What did scientists find in the asteroid Ryugu?

- (A) Gold and silver
- (B) A new planet
- (C) All five letters of DNA
- (D) Only one letter of DNA

2. What are the five letters called?

- (A) Asteroids
- (B) Nucleotides
- (C) Molecules
- (D) Nucleobases

3. Which spacecraft brought the rocks back?

- (A) A Russian rocket
- (B) NASA's Bennu mission
- (C) The International Space Station
- (D) Japan's Hayabusa2

4. In what year did the rocks come back to Earth?

- (A) 2023
- (B) 2025
- (C) 2020
- (D) 2019

5. How much rock did Hayabusa2 bring back?

- (A) 540 grams
- (B) 5.4 grams
- (C) 54 grams
- (D) 5.4 kilograms

6. Why are the Ryugu rocks special?

- (A) They were kept safe from Earth's dirt
- (B) They fell in Australia
- (C) They are very big
- (D) They glow in the dark

Answer key (try the quiz first!): 1: C 2: D 3: D 4: C 5: B 6: A

Questions to think about

MY THOUGHTS (at least 20 words):